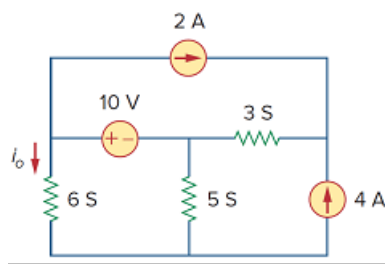


Problem

Apply nodal analysis to find i_o and the power dissipated in each resistor in the circuit of Fig. 3.64.

Figure 3.64:



Step-by-step solution

Step 1 of 11

Refer to the circuit diagram in Figure 3.64 in the textbook.

The resistors are represented by their values of conductance G with units Siemen 'S'. The relation between the resistance R and conductance G is,

$$R = \frac{1}{G}$$

The current through the resistor is,

$$\begin{aligned} i &= \frac{v}{R} \\ &= Gv \end{aligned}$$

Here, v is the voltage across the resistor.

The power consumed by the resistor is,

$$\begin{aligned} P &= \frac{v^2}{R} \\ &= Gv^2 \end{aligned}$$

(or)

The power consumed by the resistor is,

$$\begin{aligned} P &= i^2 R \\ &= \frac{i^2}{G} \end{aligned}$$

Step 2 of 11

Draw the circuit with node voltages as shown in Figure 1.

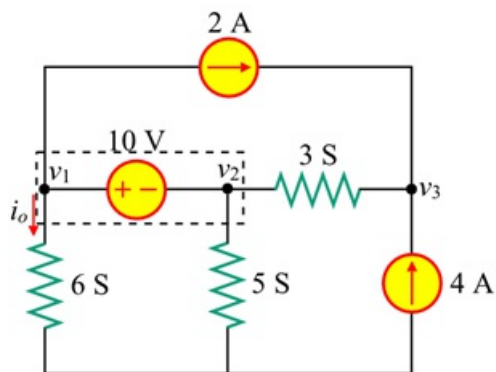


Figure 1

Step 3 of 11

From Figure 1, the nodes v_1 and v_2 form a super-node represent with dashed lines.

$$v_1 - v_2 = 10$$

$$v_1 = 10 + v_2 \dots\dots (1)$$

Apply the Kirchhoff's current law to the super node.

$$6v_1 + 2 + 5v_2 + 3(v_2 - v_3) = 0$$

$$6v_1 + 5v_2 + 3v_2 - 3v_3 = -2$$

$$6v_1 + 8v_2 - 3v_3 = -2 \dots\dots (2)$$

Step 4 of 11

Apply the Kirchhoff's current law at node 3.

$$3(v_3 - v_2) - 4 - 2 = 0$$

$$3(v_3 - v_2) = 6$$

$$v_3 - v_2 = 2$$

$$v_3 = 2 + v_2 \dots\dots (3)$$

Consider equations (1) and (3).

Substitute $(10 + v_2)$ for v_1 and $(2 + v_2)$ for v_3 in the equation (2).

$$6(10 + v_2) + 8v_2 - 3(2 + v_2) = -2$$

$$60 + 6v_2 + 8v_2 - 6 - 3v_2 = -2$$

$$(6 + 8 - 3)v_2 = -2 - 60 + 6$$

$$11v_2 = -56$$

$$v_2 = \frac{-56}{11}$$

$$= -5.09 \text{ V}$$

Step 5 of 11

Recall equation (1).

$$v_1 = 10 + v_2$$

Substitute -5.09 V for v_2 in equation (1).

$$\begin{aligned} v_1 &= 10 + (-5.09) \\ &= 10 - 5.09 \\ &= 4.91 \text{ V} \end{aligned}$$

Step 6 of 11

Recall equation (3).

$$v_3 = 2 + v_2$$

Substitute -5.09 V for v_2 in equation (3).

$$\begin{aligned} v_3 &= 2 + (-5.09) \\ &= 2 - 5.09 \\ &= -3.09 \text{ V} \end{aligned}$$

Step 7 of 11

Calculate the current through 6 S conductance.

$$\begin{aligned} i_o &= 6v_1 \\ &= 6(4.91) \\ &= 29.46 \text{ A} \end{aligned}$$

Therefore, the value of current, i_o is 29.46 A .

Step 8 of 11

Calculate the power dissipated by the 6 S conductance.

$$\begin{aligned} P_{6S} &= 6v_1^2 \\ &= (6)(4.91)^2 \\ &= (6)(24.1081) \\ &= 144.65 \text{ W} \end{aligned}$$

Therefore, the power dissipated by the 6 S conductance, P_{6S} is 144.65 W .

Step 9 of 11

Calculate the current through 3 S conductance.

$$\begin{aligned} i_{3S} &= 3(v_2 - v_3) \\ &= (3)(-5.09 - (-3.09)) \\ &= (3)(-2) \\ &= -6 \text{ A} \end{aligned}$$

Step 10 of 11

Calculate the power dissipated by the 3 S conductance.

$$\begin{aligned}P_{3\text{S}} &= \frac{i_{3\text{S}}^2}{3} \\&= \frac{(-6)^2}{3} \\&= \frac{36}{3} \\&= 12\text{ W}\end{aligned}$$

Therefore, the power dissipated by the 3 S conductance, $P_{3\text{S}}$ is $\boxed{12\text{ W}}$.

Step 11 of 11

Calculate the current through 5 S conductance.

$$\begin{aligned}i_{5\text{S}} &= 5v_2 \\&= (5)(-5.09) \\&= -25.45\text{ A}\end{aligned}$$

Calculate the power dissipated by the 5 S conductance.

$$\begin{aligned}P_{5\text{S}} &= \frac{i_{5\text{S}}^2}{5} \\&= \frac{(-25.45)^2}{5} \\&= \frac{647.7025}{5} \\&= 129.45\text{ W}\end{aligned}$$

Therefore, the power dissipated by the 5 S conductance, $P_{5\text{S}}$ is $\boxed{129.45\text{ W}}$.